

How To Use RELAYS

For Real-World Applications



Vijaykumar Sajjanar is assistant professor, Department of Electronics & Communication, BLDEA's CET, Vijayapur, Karnataka



Rajinderkumar M. Math is assistant professor, Department of Electronics & Communication, BLDEA's CET

Very often, novel and innovative projects end up remaining only academic projects, because no one is able to implement the ideas as real-world applications. Real-world applications are those that can be utilised in day-to-day life with no or very few modifications in the projects.

Projects inherently consist of three phases. First phase includes generating a system that processes; second phase verifies the system with simple sensors and actuators, and the final phase covers the system using real-world sensors and actuators.

Fig. 1 gives an idea about the three phases for controlling a fan. In phase I, a temperature sensor provides a small amount of DC voltage, which is mapped to a digital number for decision making through coding.

In phase II, these decisions drive simple actuators, such as a DC fan typically running on 5V.

In phase III, controlling of higher voltage DC/AC devices, like a ceiling fan, by means of a microcontroller (MCU) or microprocessor is completed and demonstrated. This phase is found tedious by most

people. Knowledge about relays that are used in most real-world applications from home appliances to aviation help in building impressive projects.

Relays are switches that control one circuit through another, without physical contact. Generally, these control high-power devices through low-power signals. This article provides four simple applications that can help in understanding the control of DC/AC devices to develop real-world applications.

Relays work on either electromagnet, reed or semiconductor principles. Electromagnet-controlled relays operate a mechanical switch by attracting a lever from one position to another.

Semiconductor-controlled relays use LEDs, photodiodes, SCRs and thyristors for switching a circuit on/off.

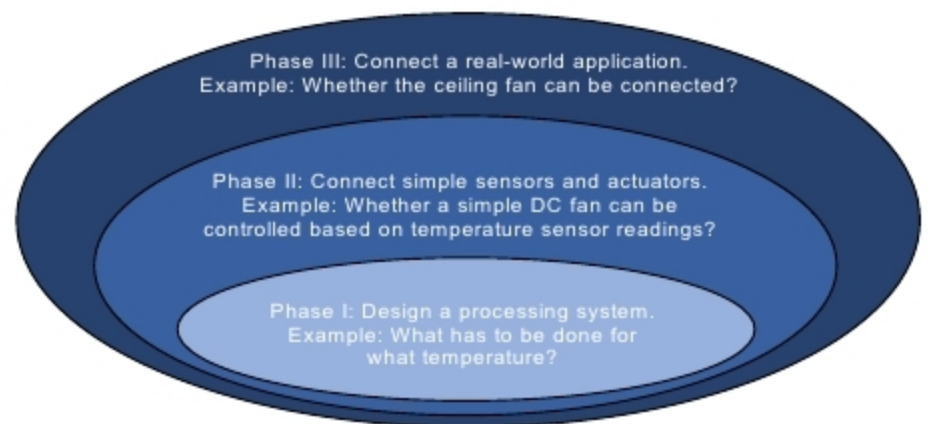


Fig. 1: Phases in fan speed control

COMPARISON BETWEEN DIFFERENT TYPES OF RELAYS

Relay	Electromagnetic relay	Reed relay	Solid-state relay
Operating principle	An electromagnet when energised moves a lever to make contact	Leads made of ferromagnetic material in a glass casing attract when subjected to magnetic field	Semiconductor devices are used to switch on/off a circuit; no physical moving parts involved
Size	Large	Small	Small
Cost	Not too costly	Costliest	High cost
Speed of operation	Slow	Fast	Fastest
Life expectancy	Less	More	Very high
Power consumption	High	Less	Less
Advantage	Very cheap, versatile; can be used with any circuit	Current requirement is very less; smaller and faster than electromagnetic relays	Very fast, less current requirement and highly durable
Disadvantage	Subjected to wear and tear	Very costly	Bound by specific voltage/current characteristics
Applications	Automobiles, aviation, industries, agriculture	Telephone exchanges, communication field	Printed circuit boards for home appliances and electrical circuits

This article provides four simple applications that can help understand the control of DC/AC devices to develop real-world applications